

PITUFFIK THULE AB, Greenland

WMO: 042020

Lat: 76.533N

Lon: 68.750W

Elev: 77

StdP: 100.41

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-35.1	-33.6	-36.1	0.1	-33.9	-34.8	0.1	-32.5	19.3	-12.5	16.0	-16.3	4.0	90	1.038

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.6	12.1	7.1	10.8	6.7	9.4	6.3	7.9	10.6	7.2	9.8	6.5	8.9	5.3	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
6.1	5.9	8.2	5.1	5.5	7.6	4.2	5.2	7.0	24.7	10.7	23.2	10.1	21.8	9.2	10.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.4	12.5	10.2	-36.4	15.1	2.4	1.7	-38.2	16.3	-39.6	17.3	-41.0	18.3	-42.7	19.6		
(5)			-36.1	8.8	2.3	1.1	-37.7	9.6	-39.0	10.3	-40.3	10.9	-42.0	11.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-10.1	-23.3	-24.4	-23.5	-15.3	-4.3	2.9	6.2	4.8	-1.5	-9.0	-14.9	-19.8		
	DBStd	12.09	5.74	6.13	6.21	5.76	4.37	3.01	2.71	2.66	3.72	5.05	5.37	6.09		
	HDD10.0	7343	1033	963	1038	759	444	213	122	163	345	590	748	924		
	HDD18.3	10380	1291	1196	1297	1009	702	462	377	421	595	848	998	1183		
	CDD10.0	4	0	0	0	0	0	1	3	1	0	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	4.0	4.4	4.4	4.0	3.6	3.3	3.6	3.8	3.6	3.8	4.7	4.8	4.6		
	PrecAvg	113	5	5	5	5	7	7	18	23	11	12	8	9		
(15) Precipitation	PrecMax	171	22	19	23	18	15	25	63	92	27	31	24	48		
	PrecMin	12	0	0	0	0	0	0	0	0	0	0	0	0		
	PrecStd	42	6	5	6	5	5	8	16	24	8	9	6	11		
	PrecStd	42	6	5	6	5	5	8	16	24	8	9	6	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.6	-3.0	-0.1	1.1	7.1	13.2	15.0	13.0	7.7	3.9	1.9	0.0		
		MCWB	-5.2	-4.2	-2.5	-0.7	4.0	7.2	8.2	7.6	3.9	1.6	0.4	-0.9		
	2%	DB	-8.9	-8.1	-6.9	-1.8	4.8	10.8	12.7	10.9	5.8	1.9	-1.9	-4.1		
		MCWB	-9.7	-9.1	-7.8	-3.1	2.2	6.3	7.4	6.5	3.1	0.3	-2.9	-5.3		
	5%	DB	-12.0	-12.4	-12.0	-4.1	3.0	9.0	11.2	9.6	4.2	0.0	-4.5	-7.9		
		MCWB	-12.6	-13.1	-12.7	-5.2	0.9	5.5	7.0	6.4	2.2	-1.1	-5.3	-8.8		
	10%	DB	-14.9	-15.6	-14.9	-6.6	1.2	7.3	10.1	8.6	3.2	-2.0	-6.9	-10.8		
		MCWB	-15.5	-16.1	-15.4	-7.6	-0.5	4.4	6.7	6.0	1.5	-2.9	-7.8	-11.5		
	0.4%	WB	-4.6	-4.2	-2.7	-0.6	4.4	7.9	8.9	8.7	4.5	2.0	0.3	-1.1		
		MCDB	-3.3	-3.2	-0.5	0.9	6.6	11.8	13.1	10.9	6.7	3.5	1.5	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-9.8	-9.3	-7.7	-3.0	2.3	6.5	8.1	7.7	3.4	0.3	-2.8	-5.3		
		MCDB	-8.9	-8.6	-6.8	-2.0	4.3	10.1	11.2	9.5	4.9	1.6	-2.0	-4.3		
	5%	WB	-12.6	-13.1	-12.6	-5.3	0.9	5.5	7.5	6.8	2.6	-1.1	-5.4	-8.7		
		MCDB	-12.0	-12.6	-11.9	-4.2	2.6	8.5	10.4	8.7	4.0	0.0	-4.5	-8.0		
	10%	WB	-15.4	-16.1	-15.3	-7.5	-0.3	4.6	6.9	6.1	1.6	-2.9	-7.7	-11.4		
		MCDB	-14.9	-15.7	-14.8	-6.5	1.1	7.2	9.6	8.0	3.1	-2.1	-7.0	-10.8		
	5% DB	MDBR	7.3	7.3	7.4	8.0	5.5	4.5	4.6	4.2	5.1	6.2	7.2	7.4		
		MCDBR	10.4	11.1	10.1	8.3	6.8	7.0	6.1	5.6	5.2	5.7	8.9	9.7		
		MCWBR	9.7	10.4	9.3	7.3	4.9	4.0	3.0	3.1	3.6	4.7	8.3	9.0		
		MCWBR	10.6	11.1	10.0	7.9	6.4	6.5	5.4	4.8	4.7	5.7	8.8	9.8		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.8	10.4	9.3	7.0	4.7	4.0	3.1	2.9	3.5	4.8	8.3	9.2		
		MCWBR	9.8	10.4	9.3	7.0	4.7	4.0	3.1	2.9	3.5	4.8	8.3	9.2		
	Clear-Sky Solar Irradiance	taub	N/A	0.145	0.220	0.267	0.284	0.303	0.299	0.284	0.247	N/A	N/A	N/A		
		taud	N/A	1.877	2.095	2.120	2.204	2.303	2.418	2.438	2.291	N/A	N/A	N/A		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	0	196	666	802	856	854	845	799	649	N/A	0	0		
		Edh at Noon	0	10	54	91	101	96	82	68	46	N/A	0	0		
(44) All-Sky Solar Radiation	RadStd	RadAvg	0.00	0.06	1.17	3.77	5.98	6.42	5.22	3.14	1.35	0.19	0.00	0.00		
		RadStd	0.00	0.01	0.05	0.18	0.39	0.66	0.42	0.39	0.12	0.03	0.00	0.00		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.98	N/A	N/A	N/A	N/A	N/A	-354	-361	N/A	N/A
(47) Regional (2 neighbors)		+0.98	N/A	N/A	N/A	N/A	N/A	-354	-361	N/A	N/A

Nomenclature: See separate page